

Capturing Concepts with 3D Scanning and Modelling

Today, aesthetics play pivotal role as a key product differentiator. 3D scanning and modelling provides designers with a shorter and smarter way to create newer concepts. Next-generation 3D laser scanners showcase high accuracy, portability, ease of use and integration with 3D modelling software tools

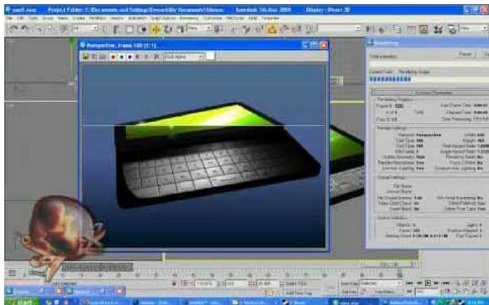
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What is the difference between HTC's Hero and Motorola's Flipout Square mobile phones? Though both are Android phones with innovative design—Hero with an angled chin and Flipout with a square shape—there is a huge difference in user experience while using the same software on a smaller screen (where you have to scroll down often to view the menu) and also in the placement of the microphone and speakers.

In order to improvise any existing product, designers need to create its 3D model by reverse engineering. This is where 3D scanning comes into play. Earlier more applicable for mechanical engineers engaged in computer-aided design, it is today very important for embedded design engineers too to understand the industrial design of the product.

According to Rahul Singh, senior designer from Longcheer, "3D measurement is a technique to analyse and measure the existing objects, which helps while developing a new concept taking an inspirational motivation from the existing objects."

A wide range of applications including robotics, aeronautics, medical electronics, gaming and biometrics require acquisition of the three-dimensional geometry of a shape to sustain competition of the aesthetics. Many of these applications require the 3D data



3D image modelling of a cellphone's keypad using the software

to be captured in real time.

Getting a low-cost, high-speed, real-time 3D scanner has always been a challenge despite the availability of different acquisition methods. Today, there are a number of portable laser scanners available in the market. Some of the latest technologies that go in today's scanners include ARM processor and digital light processing (DLP).

3D measurements: scanning and modelling

Three-dimensional measurements are carried out along all three axes and used to understand the geometry of an object for redesigning, reverse engineering, modelling of existing

products, creating concepts and more. 3D scanning helps in gathering the required data to model your imagination. To realise a concept and share your imagination with others, 3D modelling helps a great deal.

3D scanning is a process which analyses the real-world object and collects the data on its shape and appearance. The scanners are similar to cameras but the purpose of a 3D scanner is different in the way it produces point-cloud data of the surface of the geometry.

"When you move a 3D scanner above an object, it collects information about the surfaces which are not hidden. The camera collects the colour in-